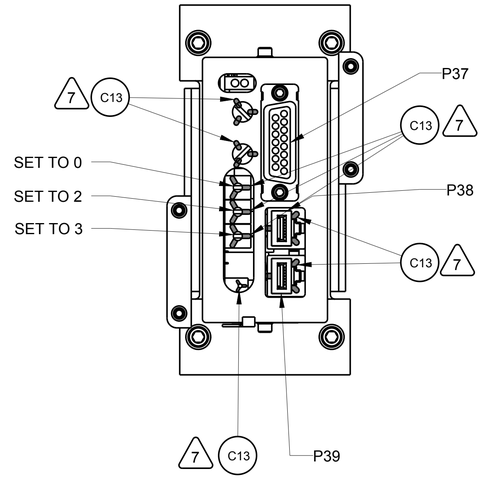
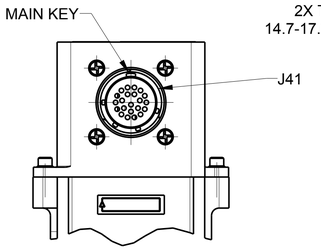
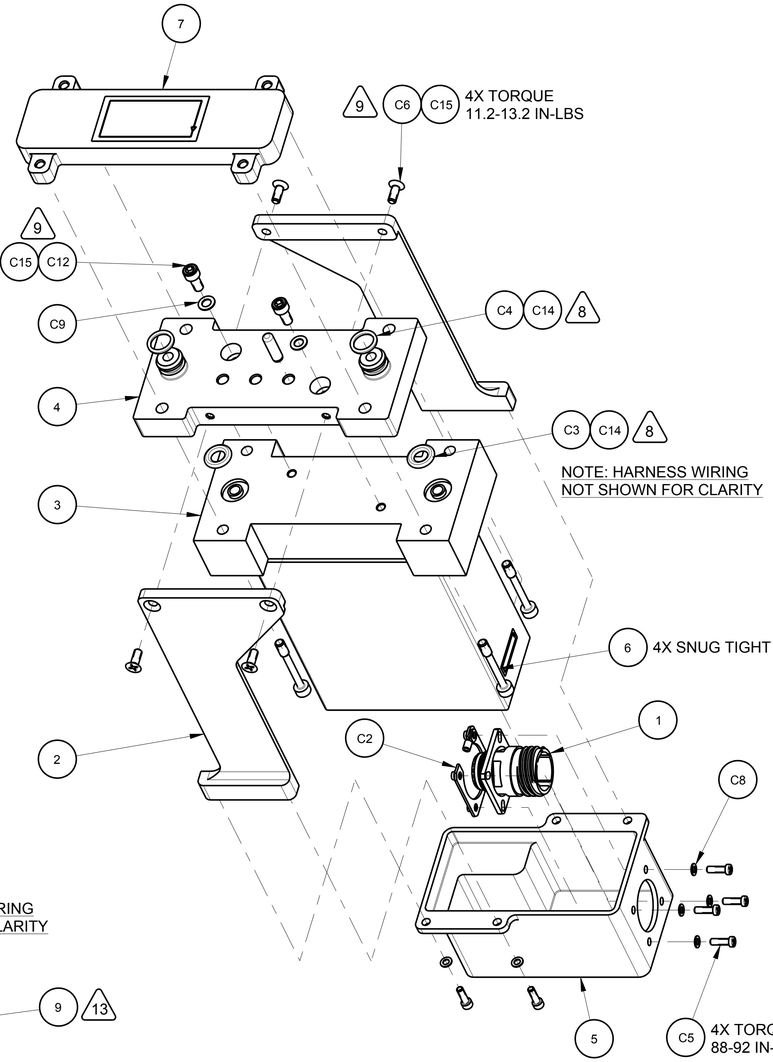
COMMERCIAL PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION
C1	1	MS16555-640	HEADLESS STRAIGHT PIN, STAINLESS STEEL .1875 OD X .625 LG, NASM16555
C2	1	M85049/95-14A	NUT PLATE, 3/4 PERIMETER, FLANGE MOUNT RECEPTACLE, SHELL SIZE C
C3	2	M83248/1-109	O-RING, 75 DUROMETER FLUOROCARBON, .299 ID X .103 W
C4	2	M83248/1-012	O-RING, 75 DUROMETER FLUOROCARBON, .364 ID X .070 W
C5	4	NAS1101E04-6	SCREW, MACHINE-FLAT FILLISTER HEAD .112-40 UNJC-3A X .375 LENGTH
C6	4	NAS1102E06-6	.138-32 UNJC-3A X .375 FLAT 100 DEG HEAD MACHINE SCREW
C7	2	NAS620C4	#4 FLAT WASHER REDUCED OUTER DIAMETER
C8	8	NAS620C4L	#4 REDUCED THICKNESS FLAT WASHER REDUCED O.D.
C9	2	NAS620C8L	#8 REDUCED THICKNESS FLAT WASHER REDUCED O.D.
C10	2	NAS1352N04-3	#4-40(.112) UNRC-3A X .188 SOCKET HEAD CAP SCREW
C11	4	NAS1352N04-6	#4-40(.112) UNRC-3A X .375 SOCKET HEAD CAP SCREW
C12	2	NAS1352N08-6	#8-32 (.164) UNRC-3A X .375 SOCKET HEAD CAP SCREW
C13	A/R	3145 GRAY	RTV ADHESIVE/SEALANT, GRAY, DOW CORNING
C14	A/R	MIL-PRF-27617	DUPONT KRYTOX 240AC TYPE III LUBRICANT
C15	A/R	222MS	THREADLOCKER, LOCTITE
C16	1	SDG32104805-009	DECAL, ELECTROSTATIC SYMBOL .5 IN X .5 IN, METAL PHOTO AND ADHESIVE (JSC)
C17	1	660-024M13N	PROTECTIVE COVER, RECEPTACLE, SHELL SIZE 13 (GLENAIR)

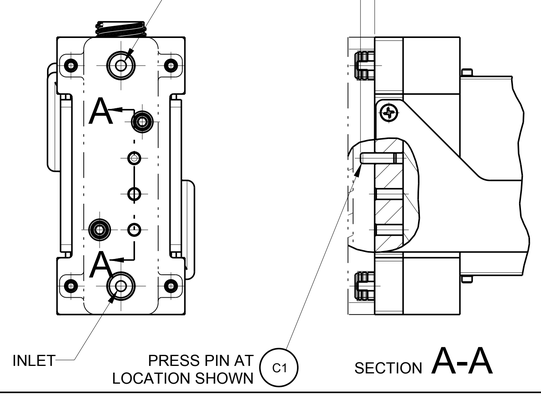


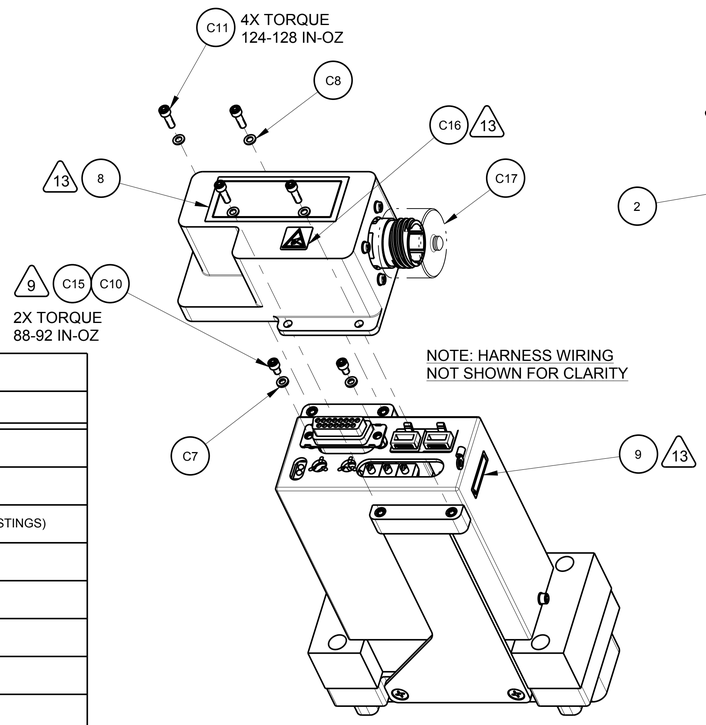


SCALE 1/1



NOTE: HARNESS WIRING NOT SHOWN FOR CLARITY





NOTE: HARNESS WIRING NOT SHOWN FOR CLARITY

DRAWING LIST

ITEM	QTY	DRAWING NUMBER	DESCRIPTION
1	1	S1101EFA1118	FUEL MASS FLOW CONTROLLER HARNESS ASSEMBLY
2	2	S1101MFA1114	MASS FLOW CONTROLLER COVER BRACKET ASSEMBLY
3	1	S1101MFA1140-008	MODIFIED HFC-D-302 0-2 SLM MASS FLOW CONTROLLER (TELEDYNE HASTINGS)
4	1	S1101MFD1112	MASS FLOW CONTROLLER MOUNTING PLATE
5	1	S1101MFD1113	MASS FLOW CONTROLLER COVER
6	4	S1101MFP1116	MASS FLOW CONTROLLER MOUNTING SCREW
7	1	S1101MFA1141	MASS FLOW CONTROLLER INTERFACE COVER ASSEMBLY
8	1	S1101MFT1143-XX	COMBO LABEL
9	1	S1101MFT1143-04	CONNECTOR ID LABEL

NOTES:

1. MATERIAL CERTIFICATION REQUIRED PER QUALITY LEVEL SPECIFIED IN WORK ORDER/SOW.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. ASSEMBLE PER APPROVED PROCESS PLAN FOR INITIAL BUILD.
4. CHEMICAL CONVERSION COAT ANY DAMAGED OR MACHINED AREAS ON ALUMINUM PARTS PER MIL-DTL-5541, TYPE I, CLASS 3.
5. CLEAN ALL SURFACES WITH ISOPROPYL ALCOHOL AND LET AIR DRY. WIPE WITH CLEAN COTTON SWABS AND/OR LINT-FREE CLOTH.
6. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
7. STAKE COMPONENTS AT DESIGNATED AREAS SHOWN. STAKING AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.1.
8. APPLY O-RING LUBRICANT PER ESTABLISHED PROCEDURES.
9. APPLY THREAD LOCKING COMPOUND PER ESTABLISHED PROCEDURES.
10. TORQUE VALUES NOTED ON THE DRAWING ARE EFFECTIVE TORQUE ONLY. RUNNING TORQUE IS REQUIRED FOR SELF LOCKING DEVICES AND BOLT FRICTION. RUNNING TORQUE MUST BE MEASURED AND ADDED TO THE EFFECTIVE TORQUE. RECORD RUNNING TORQUE AND FINAL TORQUE ON DRAWING OR ASSEMBLY PROCEDURE.
11. BEFORE STORAGE, PROTECT WITH CONTAMINATION BARRIER. USE CLEAN, NEW, ESD BAG, TOTES OR APPROVED ESD PACKAGING. MARK PART NUMBER, A DASH, AND A SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER/SOW, BY TAGGING PER SAE-AS478-35D.
12. REMOVE ALL VENDOR AFFIXED LABELS FROM ITEM 3, MASS FLOW CONTROLLER, AND TEMPORARY LABELS FROM ITEM 1 HARNESS.
13. APPLY LABEL TO CLEAN, DRY SURFACE IN LOCATION SHOWN.
14. SEE SHEET 2 FOR ADDITIONAL INFORMATION.
15. ENGINEER, OR TECHNICIAN, SHALL FUNCTIONALLY TEST COMPLETED ASSEMBLY PER ACME-TST-030.
16. ENGINEER, OR TECHNICIAN, SHALL PERFORM CONTROL TUNING PROCEDURE ON COMPLETED ASSEMBLY PER ACME-TST-041.

REV	DESCRIPTION	APPROVED
B	INCORPORATED ACME-ECO-535	AK 11/21/16 LF 11/21/16 AD 11/23/16
A	INCORPORATED ACME-ECO-458 AND ACME-ECO-512.	J.OGRIN 09/28/15 W.BOOTH 09/28/15 A.DRAKE 10/02/15 J.BRUEWER 10/14/15

REVISION HISTORY

DWG NO S1101MFA1143 | REV B | SH 1

S1101MFA0100
NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. ALL DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
X ± 0.03
XX ± 0.01
XXX ± 0.005
5. ANGULAR TOLERANCE ± 0.5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL. DO NOT REVISE WITHOUT AN APPROVED ECO.
APPROVED: S.CORN-CAIN 10/15/15
PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	K.RHODES 7/11/14	QUALITY	D.SEELEY 7/15/14
REVIEWER	G.FEDEVICH 7/16/14	MATERIALS	M.WILKINSON 7/14/14
MECHANICAL	A.DRAKE 7/11/14	PROJ LEAD	B.BOROWSKI 7/21/14
ELECTRICAL	T.GOBELI 7/11/14		
STRUCTURAL	C.WERNER 7/17/14		
SAFETY	W.BORRELLI 7/11/14		

DIMENSIONS AND TOLERANCING PER ASME Y14.5M-1994

CONTRACT NUMBER: NNC14CA02C
PROJECT CODE: S2-BT3

ZIN TECHNOLOGIES
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NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

TITLE
FUEL MASS FLOW CONTROLLER ASSEMBLY, 2.0 SLPM

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101MFA1143	B

SCALE 3/4 & NOTED | INITIAL REL DATE 7/21/14 | SHEET 1 OF 2

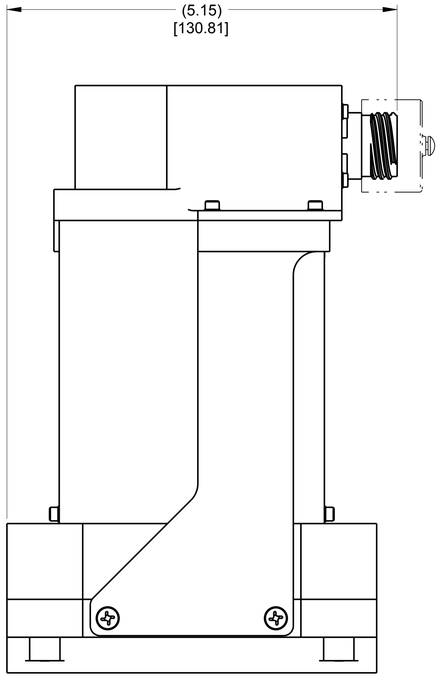
RELEASE LEVEL: RELEASED
MODEL: S1101MFA1143

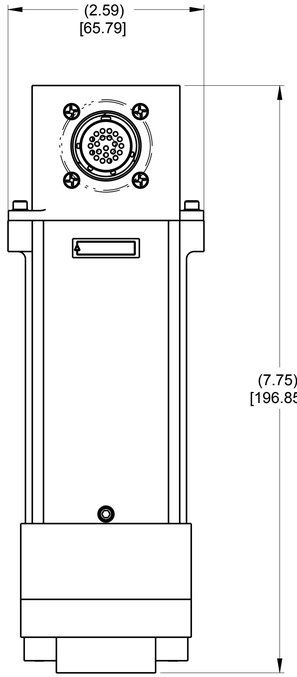
PRO/E VERSION: Wildfire 2.0

ZIN_D_2012.frm

NOTES:

1. SEE SHEET 1 FOR ADDITIONAL INFORMATION, GENERAL NOTES AND TOLERANCES.





SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101MFA1143	B
SCALE 1/1	INITIAL REL DATE 7/21/14	SHEET 2 OF 2	

ZIN_D_2012.frm

PRO/E VERSION: Wildfire 2.0

DWG NO S1101MFA1143 SH 2 REV B